

LANDSLIDES, AVALANCHES AND GLOF RISK

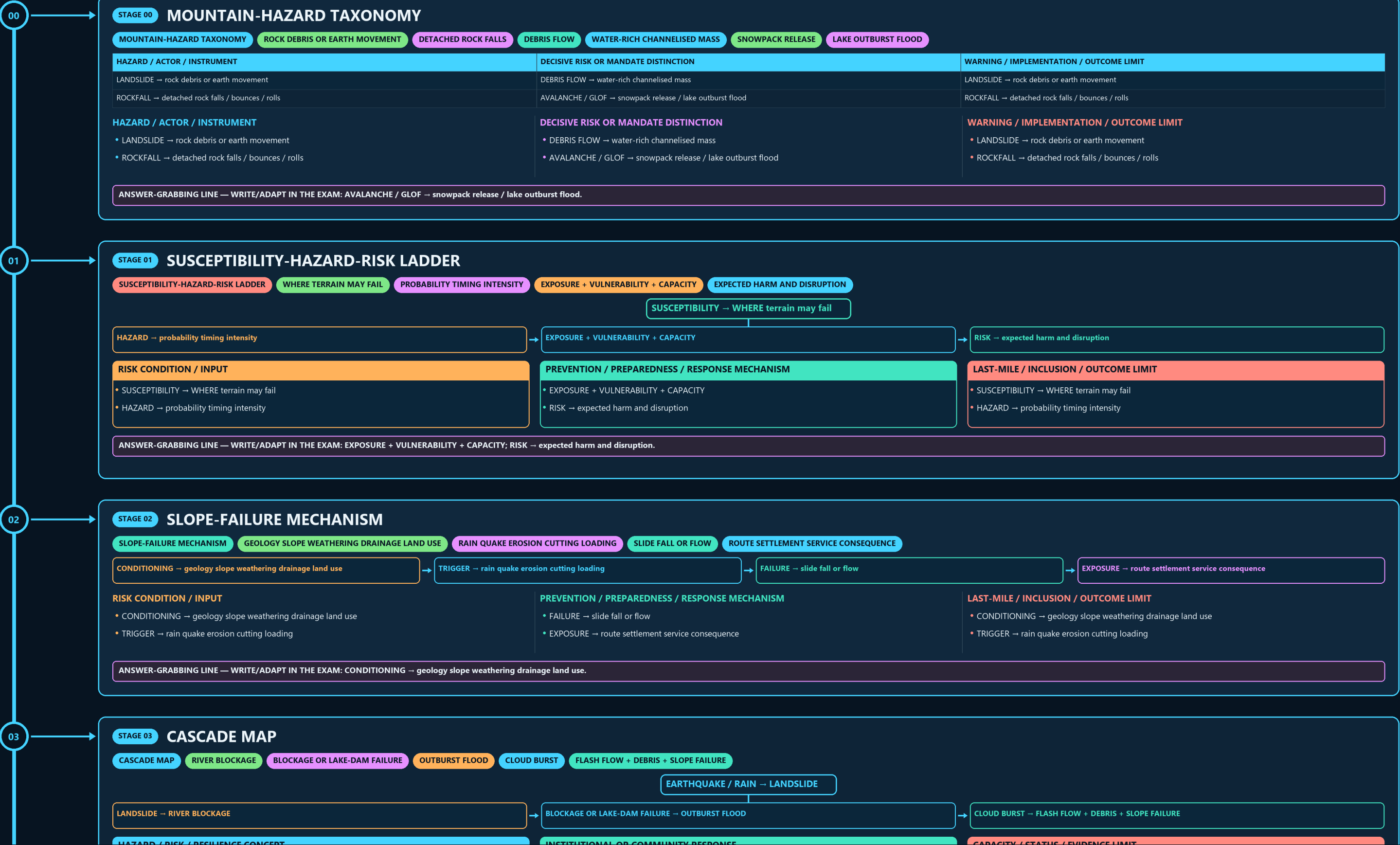
MOUNTAIN-HAZARD TAXONOMY → SUSCEPTIBILITY-HAZARD-RISK LADDER → SLOPE-FAILURE MECHANISM → CASCADE MAP → INSTITUTIONAL OWNER MATRIX → GLACIAL-LAKE MONITORING RAIL

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



06

STAGE 06 ZONATION-TO-DECISION BRIDGE

ZONATION-TO-DECISION BRIDGE

HIGHER SUSCEPTIBILITY

AVOID OR REQUIRE SITE INVESTIGATION

CONTROL LAND USE DRAINAGE CUTTING LOADING

MAP ROUTE CLOSURE EVACUATION AND RESIDUAL RISK

HIGHER SUSCEPTIBILITY / HAZARD?

AVOID OR REQUIRE SITE INVESTIGATION

CONTROL LAND USE DRAINAGE CUTTING LOADING

MAP ROUTE CLOSURE EVACUATION AND RESIDUAL RISK

HAZARD / RISK / RESILIENCE CONCEPT

- HIGHER SUSCEPTIBILITY / HAZARD?
- AVOID OR REQUIRE SITE INVESTIGATION

INSTITUTIONAL OR COMMUNITY RESPONSE

- CONTROL LAND USE DRAINAGE CUTTING LOADING
- MAP ROUTE CLOSURE EVACUATION AND RESIDUAL RISK

CAPACITY / STATUS / EVIDENCE LIMIT

- HIGHER SUSCEPTIBILITY / HAZARD?
- AVOID OR REQUIRE SITE INVESTIGATION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONTROL LAND USE DRAINAGE CUTTING LOADING; MAP ROUTE CLOSURE EVACUATION AND RESIDUAL RISK.

07

STAGE 07 SLOPE-MEASURE PORTFOLIO

SLOPE-MEASURE PORTFOLIO

SUBSURFACE DRAINAGE

CONTROLLED EXCAVATION AND TOE PROTECTION

RETAINING PROTECTION BIOENGINEERING

SITE-SPECIFIC DESIGN MAINTENANCE RESIDUAL RISK

WATER → surface / subsurface drainage

GEOMETRY → controlled excavation and toe protection

MATERIAL → retaining protection bioengineering

LIMIT → site-specific design maintenance residual risk

HAZARD / RISK / RESILIENCE CONCEPT

- WATER → surface / subsurface drainage
- GEOMETRY → controlled excavation and toe protection

INSTITUTIONAL OR COMMUNITY RESPONSE

- MATERIAL → retaining protection bioengineering
- LIMIT → site-specific design maintenance residual risk

CAPACITY / STATUS / EVIDENCE LIMIT

- WATER → surface / subsurface drainage
- GEOMETRY → controlled excavation and toe protection

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MATERIAL → retaining protection bioengineering; LIMIT → site-specific design maintenance residual risk.

08

STAGE 08 ROUTE AND SETTLEMENT RESILIENCE

ROUTE AND SETTLEMENT RESILIENCE

SITING + SPOIL

DRAINAGE CONTROL

ALTERNATIVE ACCESS + SAFE STOPPAGE

ASSEMBLY AREA + EVACUATION ROUTE

HOSPITAL POWER TELECOM WATER CONTINUITY

ALIGNMENT / SITING + SPOIL / DRAINAGE CONTROL

ALTERNATIVE ACCESS + SAFE STOPPAGE / CLOSURE

ASSEMBLY AREA + EVACUATION ROUTE

HOSPITAL POWER TELECOM WATER CONTINUITY

HAZARD / RISK / RESILIENCE CONCEPT

- ALIGNMENT / SITING + SPOIL / DRAINAGE CONTROL
- ALTERNATIVE ACCESS + SAFE STOPPAGE / CLOSURE

INSTITUTIONAL OR COMMUNITY RESPONSE

- ASSEMBLY AREA + EVACUATION ROUTE
- HOSPITAL POWER TELECOM WATER CONTINUITY

CAPACITY / STATUS / EVIDENCE LIMIT

- ALIGNMENT / SITING + SPOIL / DRAINAGE CONTROL
- ALTERNATIVE ACCESS + SAFE STOPPAGE / CLOSURE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ALIGNMENT / SITING + SPOIL / DRAINAGE CONTROL.

09

STAGE 09 WARNING-TO-ACTION CHAIN

WARNING-TO-ACTION CHAIN

COMMUNICATE UNCERTAINTY

CLOSE ROUTE

SAFE DESTINATION

VERIFY RECEIPT ACTION AND FEEDBACK

MUST REMEMBER

LANDSLIDE, AVALANCHE AND GLOF RISK IN THE HIMALAYA JOINS

MONITOR → ASSESS → FORECAST / BULLETIN

AUTHORISE → COMMUNICATE UNCERTAINTY

CLOSE ROUTE / EVACUATE → SAFE DESTINATION

VERIFY RECEIPT ACTION AND FEEDBACK

MUST REMEMBER: Landslide, avalanche and GLOF risk in the Himalaya joins slope, geology

RISK CONDITION / INPUT

- MONITOR → ASSESS → FORECAST / BULLETIN
- AUTHORISE → COMMUNICATE UNCERTAINTY

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM

- CLOSE ROUTE / EVACUATE → SAFE DESTINATION
- VERIFY RECEIPT ACTION AND FEEDBACK

LAST-MILE / INCLUSION / OUTCOME LIMIT

- MUST REMEMBER: Landslide, avalanche and GLOF risk in the Himalaya joins slope, geology,...
- MONITOR → ASSESS → FORECAST / BULLETIN

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Landslide, avalanche and GLOF risk in the Himalaya joins slope, geology,.

09

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- MONITOR → ASSESS → FORECAST / BULLETIN

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Landslide, avalanche and GLOF risk in the Himalaya joins slope, geology,.

10

STAGE 10 DEVELOPMENT-RISK BALANCE

DEVELOPMENT-RISK BALANCE

CONNECTIVITY TOURISM HYDROPOWER LIVELIHOODS

SLOPE DRAINAGE GLACIAL-LAKE AND DOWNSTREAM RISK

RISK-INFORMED SITING NOT BLANKET CLAIMS

CLOSE DISTINCTION

SUSCEPTIBILITY IS NOT PROBABILITY; TRIGGER IS NOT ROOT VULNERABILITY

HAZARD / ACTOR / INSTRUMENT

CONNECTIVITY TOURISM HYDROPOWER LIVELIHOODS

SLOPE DRAINAGE GLACIAL-LAKE AND DOWNSTREAM RISK

DECISIVE RISK OR MANDATE DISTINCTION

VERDICT → RISK-INFORMED SITING NOT BLANKET CLAIMS

CLOSE DISTINCTION: Susceptibility is not probability; trigger is not root vulnerability;...

WARNING / IMPLEMENTATION / OUTCOME LIMIT

CONNECTIVITY TOURISM HYDROPOWER LIVELIHOODS

SLOPE DRAINAGE GLACIAL-LAKE AND DOWNSTREAM RISK

HAZARD / ACTOR / INSTRUMENT

- CONNECTIVITY TOURISM HYDROPOWER LIVELIHOODS
- SLOPE DRAINAGE GLACIAL-LAKE AND DOWNSTREAM RISK

DECISIVE RISK OR MANDATE DISTINCTION

- VERDICT → RISK-INFORMED SITING NOT BLANKET CLAIMS
- CLOSE DISTINCTION: Susceptibility is not probability; trigger is not root vulnerability;...

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- CONNECTIVITY TOURISM HYDROPOWER LIVELIHOODS
- SLOPE DRAINAGE GLACIAL-LAKE AND DOWNSTREAM RISK

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Susceptibility is not probability; trigger is not root vulnerability,.

11

STAGE 11 MOUNTAIN-RISK ANSWER SPINE

MOUNTAIN-RISK ANSWER SPINE

CLASSIFY PROCESS

SEPARATE CONDITION FROM TRIGGER

DISTINGUISH SUSCEPTIBILITY HAZARD AND RISK

ASSIGN GSI DGRE NRSC CWC AND LOCAL ACTION

COMBINE ZONING DRAINAGE ROUTES WARNING EVACUATION + LIMITS

EVIDENCE LIMIT

SOURCE EVENT DATE, LOCATION, RAINFALL

CLASSIFY PROCESS → SEPARATE CONDITION FROM TRIGGER

DISTINGUISH SUSCEPTIBILITY HAZARD AND RISK

ASSIGN GSI DGRE NRSC CWC AND LOCAL ACTION

COMBINE ZONING DRAINAGE ROUTES WARNING EVACUATION + LIMITS

EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Source event date, location, rainfall

HAZARD / RISK / RESILIENCE CONCEPT

- CLASSIFY PROCESS → SEPARATE CONDITION FROM TRIGGER
- DISTINGUISH SUSCEPTIBILITY HAZARD AND RISK

INSTITUTIONAL OR COMMUNITY RESPONSE

- ASSIGN GSI DGRE NRSC CWC AND LOCAL ACTION
- COMBINE ZONING DRAINAGE ROUTES WARNING EVACUATION + LIMITS

CAPACITY / STATUS / EVIDENCE LIMIT

- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Source event date, location, rainfall,...
- CLASSIFY PROCESS → SEPARATE CONDITION FROM TRIGGER

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Source event date, location, rainfall,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Source event date, location, rainfall,.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

NDMA'S EXPLICIT NON-ISSUANCE OF CLOUDBURST-SPECIFIC GUIDELINES

STATE-LEVEL RESPONSIBILITY FOR AREA-SPECIFIC MEASURES

AVALANCHE WARNING SITS WITH A DEFENCE-RESEARCH INSTITUTION, NOT

MINING-LINKED LANDSLIDES AS A DISTINCT SUB-CATEGORY

THE UTTARAKHAND 2013 DISASTER AS A DOCUMENTED MULTI-CLOUDBURST

CLOUDBURST'S NEAR-IRREDUCIBLE PREDICTION GAP

VISIONIAS STATES

OPTIONAL RISK / RESILIENCE DEPTH

- NDMA's explicit non-issuance of cloudburst-specific guidelines
- State-level responsibility for area-specific measures : landslide
- CAUTION: Avalanche warning sits with a defence-research institution, not
- Mining-linked landslides as a distinct sub-category : VisionIAS's

MANDATE, WARNING OR OUTCOME LIMIT

- The Uttarakhand 2013 disaster as a documented multi-cloudburst,
- CAUTION: Cloudburst's near-irreducible prediction gap : VisionIAS states
- ISRO's Himalayan glacial-lake monitoring role (PDF p. 79) is the
- CAUTION: Avalanche prediction shares cloudburst's short-lead-time

COMPARATIVE HAZARD / GOVERNANCE NUANCE

- CAUTION: Himalayan/hill-state populations dependent on tourism and
- CAUTION: Hydropower/infrastructure development versus glacial-risk
- CAUTION: Terrace farming versus Jhumming (shifting cultivation) in the
- NOT: The "landslide vulnerability zones are mapped, therefore risk is

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.